

Math 196 Lecture Notes: Week 5

Miles Kretschmer

April 2026

0.1 Review of Last Week's Topics

0.1.1 Bases and Dimension

If $V \subset \mathbb{R}^n$ is a linear subspace, then a *basis* for V is a list of vectors $v_1, \dots, v_r$ such that

1. $v_1, \dots, v_r$ are linearly independent
2. $\text{span}\{v_1, \dots, v_r\} = V$

We saw that any two bases for a subspace must have the same size, and so we defined the *dimension* $\dim(V)$ as the size of any basis. The dimension is a number between 0 and n ; the only zero dimensional subspace is $\{0\}$, the only n dimensional subspace is $\mathbb{R}^n$ itself. We also saw a method to compute a basis for subspaces presented as $\text{im}(A)$ or $\text{ker}(A)$, for A a matrix.

A consequence of this was

Theorem 1 (Rank-Nullity). *If A is $m \times n$, so $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, A has n columns, then $\dim(\text{im}(A)) + \dim(\text{ker}(A)) = n$.*

0.1.2 Bases as Coordinate Systems

Any basis $v_1, \dots, v_r$ allows us to describe elements of v using r -coordinates $x_1, \dots, x_r$, where $v = x_1v_1 + \dots + x_rv_r$. Thus, $\dim(V)$ is the number of coordinates needed to describe an element of V . In particular different bases for $\mathbb{R}^n$ give us different coordinate systems on $\mathbb{R}^n$. If $v_1, \dots, v_n$ is a basis $\mathfrak{B}$, and

$$S = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

then the coordinates of v with respect to $\mathfrak{B}$ are computed using the formula

$$[v]_{\mathfrak{B}} = S^{-1}v$$

1 Linear Transformations in Different Coordinate Systems

Not only can we find representations of vectors in the coordinate system given by a basis, we can also find representations of linear maps. Suppose that $v_1, \dots, v_n$ is a basis of $\mathbb{R}^n$, and $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation. We know how to find a matrix representation of T with respect to the standard basis, but we can also find one with respect to this new basis.

For each i , we consider the representation of $T(v_i)$ in the coordinate system given by the basis $\mathfrak{B} = (v_1, \dots, v_n)$, $[T(v_i)]_{\mathfrak{B}}$. These vectors form the columns of an $n \times n$ matrix C . This matrix is a representation

of T in the following way. Given any vector v , we can compute $T(v)$ by first taking the coordinate vector $[v]_{\mathfrak{B}}$, applying C to it to get a new coordinate vector

$$C[v]_{\mathfrak{B}} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

and then translating these coordinates into the vector they represent,

$$T(v) = y_1 v_1 + \cdots + y_n v_n$$

Thus C allows us to compute Tv while staying in the $\mathfrak{B}$ coordinate system. Another way to state this is that the matrix C satisfies the equation

$$[T(v)]_{\mathfrak{B}} = C[v]_{\mathfrak{B}}$$

for any vector v .

Example 1. Let T be the linear transformation of $\mathbb{R}^2$ given by a quarter turn, rotation through $\pi/2$. Find the matrix representing T in the basis $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

We need to find the coordinates of

$$T(v_1) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, T(v_2) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

with respect to the basis. This is equivalent to applying the matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1/3 & -2/3 \\ -2/3 & 1/3 \end{bmatrix}$$

(We computed this inverse two weeks ago.) We compute

$$[T(v_1)]_{\mathfrak{B}} = \begin{bmatrix} 1/3 & -2/3 \\ -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -4/3 \\ 5/3 \end{bmatrix}$$

$$[T(v_2)]_{\mathfrak{B}} = \begin{bmatrix} 1/3 & -2/3 \\ -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -5/3 \\ 4/3 \end{bmatrix}$$

Thus the matrix we want is

$$\begin{bmatrix} -4/3 & -5/3 \\ 5/3 & 4/3 \end{bmatrix}$$

If A is the matrix representation of T in the standard basis, we can turn the above process into a formula for the representation of T in a new basis. Again let S be the matrix whose columns are the new basis vectors $v_1, \dots, v_n$. Then the columns of C are obtained by applying T to the columns of S , then applying S^{-1} . In other words, $C = S^{-1}AS$. We can also see this using the equation

$$[T(v)]_{\mathfrak{B}} = C[v]_{\mathfrak{B}}$$

This comes out to

$$S^{-1}AS = CS^{-1}$$

and so

$$S^{-1}AS = C$$

Matrices A, C with $C = S^{-1}AS$, equivalently $A = SCS^{-1}$ are called *similar*. We can say now that this means they represent the same transformation with respect to different choices of basis.

Given a transformation T , it is often easier to answer questions about it after choosing a coordinate system so that the associated matrix representation is as simple as possible. This is the idea behind the last main topic of the course, diagonalization.

Example 2. Let T be the transformation given by reflection across the line $y = -x$. On the first homework, we saw that this is given by the matrix

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Consider the basis for $\mathbb{R}^2$ consisting of the vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

We then have $T(v_1) = -v_1$, $T(v_2) = v_2$, so in this coordinate system, T is represented by the matrix

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Any reflection matrix is similar to this one, for a similar reason.

1.1 Abstract Vector Spaces

To apply tools from linear algebra, we often consider objects which are not themselves vectors, but which can be represented as vectors. You saw an example of this on the first homework, with a polynomial represented as a vector of coefficients. With a generalization of the notion of a linear subspace, and our view of bases as coordinate systems, we can build a convenient language to talk about this. We will not use this formalism much in this course, but I do think it is worth seeing, because it underlies much of the applicability of linear algebra to myriad problems in mathematics, such as the study of differential equations.

Definition. A *vector space* is a set V equipped with two operations:

1. Addition: given $v, w \in V$, we can define $v + w \in V$.
2. Scalar Multiplication: given $v \in V$ and $k \in \mathbb{R}$, we can define $kv \in V$.

We also ask for an element $0 \in V$. These operations should satisfy the rules of vector algebra that we know from $\mathbb{R}^n$, in particular:

$$a(v + w) = av + aw, (a + b)v = av + bv$$

$$a(bv) = (ab)v$$

$$a \cdot 0 = 0, v + 0 = v$$

In any vector space, the notions of linear combinations, span, linear subspace, linear independence, bases, and dimension all make sense, as we have defined them for subspaces of $\mathbb{R}^n$.

Example 3. Any subspace $V \subset \mathbb{R}^n$ is a vector space.

Example 4. Let P be the collection of polynomials in one variable t . We can add polynomials, and multiply them by constants, so P is a vector space. Let P_3 be the collection of all polynomials of degree at most 3. This contains 0, and is closed under addition and scalar multiplication, so is a subspace of P . Any polynomial of degree at most 3 can be uniquely written as a linear combination of $1, t, t^2, t^3$:

$$f(t) = a_3t^3 + a_2t^2 + a_1t + a_0$$

so this is a basis for P_3 , showing that $\dim(P_3) = 4$. The vector space P has infinite dimension; a basis for it is the infinite set $\{1, t, t^2, t^3, t^4, \dots\}$.

If we have a (finite) basis $u_1, \dots, u_n$ for V a vector space, we can put a coordinate system on V , as we did with subspaces before. In particular, given $v \in V$

$$[v]_{\mathfrak{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

where $v = x_1 u_1 + \dots + x_n u_n$.

Example 5. With respect to the basis $\mathfrak{B} = (1, t, t^2, t^3)$, the polynomial $f(t) = t^3 - 2t - 1$ has coordinates

$$[f]_{\mathfrak{B}} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

1.1.1 Linear Maps Between Abstract Vector Spaces

The notion of a linear transformation also makes sense for abstract vector spaces. Letting V and W be vector spaces, a function $f : V \rightarrow W$ is linear if

1. $f(v_1 + v_2) = f(v_1) + f(v_2)$
2. $f(k \cdot v) = k \cdot f(v)$

An invertible linear function is called an *isomorphism* of vector spaces. If there is an invertible linear function $V \rightarrow W$, we say that V and W are *isomorphic*.

Remark. The word *isomorphic* means essentially “same shape.” One can think of two isomorphic vector spaces as having the same structure, with different names for the objects. The isomorphism f translates an object in V into an object in W .

Example 6. Any choice of basis $v_1, \dots, v_n$ for V yields an isomorphism

$$F : \mathbb{R}^n \rightarrow V$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \mapsto x_1 v_1 + \dots + x_n v_n$$

Specifically, we have an isomorphism

$$F : \mathbb{R}^4 \rightarrow P_3$$

$$\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto x_3 t^3 + x_2 t^2 + x_1 t + x_0$$

If $F : \mathbb{R}^n \rightarrow V$ is an isomorphism coming from a basis $\mathfrak{B}$, then $[v]_{\mathfrak{B}} = F^{-1}v$.

Any isomorphism between two vector spaces takes a basis of one to a basis of the other. This means in particular that $\dim V = \dim W$. This gives us another way of seeing why invertible matrices are always square.

Given bases $\mathfrak{B} = (v_1, \dots, v_n)$ for V , $\mathfrak{C} = (w_1, \dots, w_m)$ for W , we can represent a linear function by a matrix in the following sense. We want a matrix A so that

$$[T(v)]_{\mathfrak{C}} = A[v]_{\mathfrak{B}}$$

We can compute this matrix in much the same way as before. For each v_i , we compute $[T(v_i)]_{\mathfrak{C}}$. These vectors then form the columns of our matrix. If $V = W$ and we use the same basis $\mathfrak{B}$ on both sides, then this is completely analogous to the earlier construction.

Example 7. Consider the map $D : P_3 \rightarrow P_3$ taking a polynomial $f(t)$ to its derivative $f'(t)$. This is a linear map. To compute its matrix representation in the basis $1, t, t^2, t^3$, we compute

$$D(1) = 0$$

$$D(t) = 1$$

$$D(t^2) = 2t$$

$$D(t^3) = 3t^2$$

This give us a matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If we use a different basis for P_3 , we get a different matrix representation. In particular, we could choose instead $1, t, t^2/2, t^3/6$, in which case we get the matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

From this matrix representation, we can see that $\text{rk}(D) = 3$, $\dim(\ker(D)) = 1$.

Example 8. The map $T(f) = (t - 2) \cdot f$ is a linear transformation from P_2 to P_3 . In the basis $1, t, t^2$ for P_2 and $1, t, t^2, t^3$ for P_3 , this is represented by the matrix

$$\begin{bmatrix} -2 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

2 Dot Products and Orthogonality in Higher Dimensions

2.1 The Dot Product and Geometry in the Plane

In this section of the course, we will discuss orthogonal projection in higher dimensions, and other associated phenomena. As a warm up, we start by reviewing the geometric significance of the dot product in the plane $\mathbb{R}^2$.

2.1.1 Length of vectors

Using the Pythagorean theorem, we can compute the length of a vector $|v|$. If

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

then [draw picture] $|v|^2 = x^2 + y^2 = v \cdot v$, and so $|v| = \sqrt{v \cdot v}$.

2.1.2 Orthogonality

Previously, we said that vectors v and w are at a right angle if $v \cdot w = 0$. One way to see this is that if

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

then the vectors w with $v \cdot w = 0$ are exactly the scalar multiples of

$$\begin{bmatrix} -y \\ x \end{bmatrix}$$

(at least if $v \neq 0$) which we know is v rotated by a right angle. [Draw pictures] There is another way to see this, which works in higher dimensions as well. We first carry length into this setting.

2.2 Length and Angle in Higher Dimensions

Given a vector $v \in \mathbb{R}^n$, the Pythagorean theorem can be repeatedly applied to compute its length $|v|$: if

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

then

$$|v|^2 = x_1^2 + \cdots + x_n^2 = v \cdot v$$

Thus $|v| = \sqrt{v \cdot v}$, as in the plane. When are two vectors in $\mathbb{R}^n$ orthogonal? Well, this will happen when they form the legs of a right triangle. [Draw picture]. This means that

$$|v + w|^2 = |v|^2 + |w|^2$$

Expanding these in terms of dot products, we get

$$(v + w) \cdot (v + w) = v \cdot v + w \cdot w$$

The left side expands as

$$v \cdot v + 2(v \cdot w) + w \cdot w$$

so the vectors are orthogonal exactly when $v \cdot w = 0$, as is the case in $\mathbb{R}^2$.

2.2.1 Angle

Even if v and w are not orthogonal, their dot product still tells us about their angle with respect to each other. Consider the triangle with sides $v, w, v - w$. [Draw picture, with labeled angle θ] The *law of cosines* tells us that

$$|v - w|^2 = |v|^2 + |w|^2 - 2|v||w|\cos(\theta)$$

This is like the Pythagorean theorem with a correction for the angle. Expanding in terms of dot products, we get

$$v \cdot v - 2v \cdot w + w \cdot w = v \cdot v + w \cdot w - 2|v||w|\cos(\theta)$$

and so

$$v \cdot w = |v||w|\cos(\theta)$$

This gives us a geometric interpretation of the dot product. Vectors pointing in the same direction have dot product the product of their lengths, and vectors pointing in opposite directions have the negative of this, for example. If v and w are nonzero, then we can solve for θ :

$$\theta = \cos^{-1} \left(\frac{v \cdot w}{|v||w|} \right)$$

Example 9. What is the angle between $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$? We have $|u| = |v| = \sqrt{5}$, $u \cdot v = 1 \cdot 1 + 2 \cdot (-2) = -3$. Thus

$$\theta = \cos^{-1}(-3/5)$$

which is somewhere between $2\pi/3$ and $5\pi/6$.

Remark. Because $|\cos(\theta)| \leq 1$, we have the inequality

$$|v \cdot w| \leq |v||w|$$

Squaring both sides, we get

$$(v \cdot w)^2 \leq |v|^2 |w|^2$$

Expanding this in terms of the components of v and w , we have

$$\left[\sum_i x_i y_i \right]^2 \leq \left[\sum_i x_i^2 \right] \left[\sum_i y_i^2 \right]$$

This is called the *Cauchy-Schwarz inequality*. It is often useful to take $y_i = 1$ for all i , so the inequality becomes

$$\left[\sum_i x_i \right]^2 \leq n \sum_i x_i^2$$

2.2.2 Summary

For $v \in \mathbb{R}^n$, its length is computed as $|v| = \sqrt{v \cdot v}$. Two vectors v and w are orthogonal exactly when $v \cdot w = 0$. If v and w are both nonzero, then $|v||w| \neq 0$, and we can calculate the angle between v and w using

$$\cos(\theta) = \frac{v \cdot w}{|v||w|}$$

One could take these formulas as definitions of length, angle, and orthogonality in higher dimensions; this is the form in which we will work with these concepts. The preceding discussion was intended to motivate these definitions, by showing that they come from plane geometry.

With these concepts fixed, we can introduce some terminology that will be useful later: A *unit vector* is a vector u with $|u| = 1$. Given any nonzero vector v , we can *normalize* it to get a unit vector pointing in the same direction

$$u = \frac{1}{|v|} \cdot v$$

2.3 Orthonormal Sets of Vectors

A collection of vectors $u_1, \dots, u_r \in \mathbb{R}^n$ is said to be *orthonormal* if

1. each u_i is a unit vector: $|u_i| = 1$.
2. for $i \neq j$, u_i is orthogonal to u_j : $u_i \cdot u_j = 0$.

We can summarize this by saying that $u_i \cdot u_j = 1$ if $i = j$, and 0 otherwise.

Example 10. The standard basis $e_1, \dots, e_n$ is orthonormal.

Example 11. The columns of a rotation matrix

$$\begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$

are orthonormal.

Proposition. *An orthonormal set is linearly independent.*

Proof. The proof of this uses a generally useful property of orthonormal sets, given a linear combination

$$x_1 u_1 + \cdots + x_r u_r$$

we can extract the coefficient x_i by taking the dot product with u_i .

$$u_i \cdot (x_1 u_1 + \cdots + x_r u_r) = x_1 (u_i \cdot u_1) + \cdots + x_r (u_i \cdot u_r)$$

Because $u_1, \dots, u_r$ is orthonormal, all of these are zero except $x_i (u_i \cdot u_i) = x_i$. In particular, if we have a linear relation

$$0 = x_1 u_1 + \cdots + x_r u_r$$

then dotting with each u_i , we see that $x_1 = x_2 = \cdots = x_r = 0$. □

We say $u_1, \dots, u_r$ is a *orthonormal basis* of a subspace V if $\text{span}\{u_1, \dots, u_r\} = V$. In particular, if $u_1, \dots, u_n \in \mathbb{R}^n$ is an orthonormal set of vectors, it is an orthonormal basis of $\mathbb{R}^n$. Given any vector v , the coordinates of v in this basis are the numbers $x_i = v \cdot u_i$.

2.4 Orthogonal Projection onto a Subspace

Previously, we saw how to compute the orthogonal projection of a vector onto a line in the plane. This is like the shadow that the vector casts on the line. [Picture] We can consider a similar operation in higher dimension space: projection onto a linear subspace.

First of all, given a vector v and a subspace W , we say that v is *orthogonal to W* if $v \cdot w = 0$ for all $w \in W$.

Example 12. In $\mathbb{R}^3$, e_3 is orthogonal to the xy -plane:

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = 0$$

Proposition. *If $w_1, \dots, w_r$ is a basis for W , then v is orthogonal to W if and only if v is orthogonal to each w_i .*

Proof. Any $w \in W$ can be written $x_1 w_1 + \cdots + x_r w_r$. If v is orthogonal to each w_i , then

$$v \cdot w = \sum_i x_i (v \cdot w_i) = 0$$

Thus v is orthogonal to any vector in w . □

Definition. Given a subspace $W \subset \mathbb{R}^n$, its *orthogonal complement* $W^\perp$ is the set of all vectors v orthogonal to W .

Example 13. The orthogonal complement of a line in $\mathbb{R}^3$ is a plane, the orthogonal complement of a plane is a line. [Draw pictures]

Proposition. $W^\perp$ is a linear subspace.

Proof. This will be on the homework. □

We will see that any vector $v \in \mathbb{R}^n$ can be decomposed in a unique way as $v = v_1 + v_2$, where $v_1 \in W$, and $v_2 \in W^\perp$. The function sending v to its component in W is proj_W , orthogonal projection onto W . We will also see that this is linear.

2.4.1 Computing Orthogonal Projection Using an Orthonormal Basis

If we happen to have an orthonormal basis $u_1, \dots, u_r$ for W , then we can use this to compute $\text{proj}_W(v)$. Next week, we will study a method of constructing orthonormal bases, which will allow us to see that our definition of $\text{proj}_W(v)$ makes sense, and allows us to compute it.

Consider the vector

$$v_1 = \sum_i (v \cdot u_i) u_i$$

This is a linear combination of elements of W , so is in W . Moreover, because $u_1, \dots, u_r$ is orthonormal, we have

$$v_1 \cdot u_i = v \cdot u_i$$

for each i . Thus, if we set $v_2 = v - v_1$, we have $v_2 \cdot u_i = 0$ for each i , so $v_2 \in W^\perp$. Thus $v = v_1 + v_2$ is a decomposition of v into an element of W and an element of $W^\perp$, and so we can say that $v_1 = \text{proj}_W(v)$ in this case.

Example 14. What is the orthogonal projection of

$$w = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

onto the plane

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}?$$

Let v_1, v_2 be the vectors spanning the plane, and note that v_1 and v_2 are orthogonal, $v_1 \cdot v_2 = 1 + 1 - 2 = 0$. Thus if we normalize these vectors, we get an orthogonal basis for the plane. Define

$$u_1 = \frac{1}{|v_1|} v_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$u_2 = \frac{1}{|v_2|} v_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Now our projection is the vector $(w \cdot u_1)u_1 + (w \cdot u_2)u_2$. The coefficients are

$$w \cdot u_1 = \frac{1 + 2 - 2}{\sqrt{6}} = \frac{1}{\sqrt{6}}$$

$$w \cdot u_2 = \frac{1 + 2 + 1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

We conclude that the projection is

$$\frac{1}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + \frac{4}{3} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \\ -1 \end{bmatrix}$$